

DATATHON: ASSESSMENT SHEET

(Appendix B, refers Para 8.5 of General Instructions, Datathon-2026) · **Self-assessment of the submission before dispatch**

Participant Information

- Participant's Rank & Name : [Rank] Arjun Jayadev
- Theme : 6.2 Global Conflicts - Impact on Supply Chains (Project DARKWATER)
- Date of Assessment : 23 Sep 2026
- Documents assessed : Commander's Edition (22 pp, main submission) with Full Technical Report (84 pp) as annex

Scoring Criteria

Para	Criterion	Max	Marks	Evidence in the submission	What would lift the mark
4.	DATA ANALYTICS (40 points)		38 / 40		
4.1	Ability to understand and define the problem	8	8	Core question ('do ships leave or stop identifying themselves?') answers the theme text on AIS dead zones and the identification crisis in the Black Sea / Persian Gulf; aim, objectives and seven testable hypotheses stated (Ch 1).	-
4.2	Quality of data preprocessing and cleaning	8	7	Cleaning log for both files; confidence filter; MMSI validity; 67 possible feed-outage passes isolated; maritime pseudo-countries separated; fully repeatable pipeline (two runs give identical outputs).	Bring the forensic-screen table (unmatched-with-MMSI, all-dark passes) into the short edition.
4.3	Selection of appropriate analytical techniques	8	8	Each technique mapped to a commander's question (Annex B): change-points, survival analysis, hot-spots, clustering, difference-in-differences, event study, spatially honest machine-learning validation.	-
4.4	Depth and accuracy of analysis	8	7	Findings replicated under 8 specifications, dose-response, DiD (+5.7 pts/day, p = 0.008), stasis, out-of-sample oil-price check; failed or non-replicating claims dropped.	SAR window is 14 days; presence-vs-identity rests on two passes; defence-budget figure to be verified.
4.5	Use of relevant visualizations and insights	8	8	40+ charts and 11 infographics; every chart carries a one-line 'What this shows'; theatre map, four-signature and centre-of-gravity graphics.	-
5.	TECHNICAL SKILLS (25 points)		21 / 25		
5.1	Proficiency in programme languages / platform / software	5	3	Advanced Python pipeline (11 stages). Power BI star schema, DAX and build guide prepared, but the .pbix is not yet built.	Build and submit the .pbix (mandatory, Para 8.2) and add its screenshots: +2.
5.2	Utilization of libraries and frameworks	7	6	pandas, NumPy, SciPy, statsmodels, scikit-learn, ruptures, Matplotlib/Basemap, headless-browser reporting.	Power BI visuals and DAX in the dashboard would confirm this mark.
5.3	Efficient data manipulation and transformation	8	7	Haversine spatial joins, scene/pass aggregation, weekly panels, star-schema export (FactConflictWeekly, FactVesselDetections).	Show Power Query steps in the .pbix: +1.
5.4	Application of statistical methods	5	5	Wilson intervals, chi-square, Mann-Whitney, bootstrap, permutation tests, Kaplan-Meier/Cox, ARIMA back-test, Granger.	-
6.	CREATIVITY AND INNOVATION (20 points)		19 / 20		
6.1	Uniqueness and originality of the approach	6	6	Maritime Picture Assurance; Strategic Dark Ratio; presence vs identity; four signatures.	-
6.2	Creative problem-solving techniques	6	6	Satellite revisits used to detect held ships; survival curves turned into a 35-45 day stock rule; I&W tested out-of-sample.	-
6.3	Innovative use of data and methods	8	7	Fuses CDM conflict + SAR/AIS data with open oil-price, rupee, energy and port data.	Future work: a longer SAR record with a pre-war baseline (not achievable before 30 Sep).
7.	COMMUNICATION (15 points)		14 / 15		

7.1	Clarity and organization of the report / presentation	5	5	Decision brief, story arc (Prologue, Parts I-VIII, Epilogue), 'So what' boxes, contents and annexes.	-
7.2	Ability to convey complex ideas concisely	5	4	22-page edition with plain-words glossary; executive summary is still number-dense.	Trim the executive summary to five numbers: +1.
7.3	Engaging and informative data storytelling	5	5	Single narrative from the storm on land to the Armed Forces; each chapter links to the previous.	-

TOTAL: 92 / 100 as the submission stands today | **96 / 100** achievable by 30 Sep once actions 1-5 below are closed.

8. Overall Impact - 8.1 Rate the overall impact on a scale of 1-10 : **9 / 10** (a decision brief with a costed, no-new-money first phase, a validated early warning and quantified implications for all three Services).

Sub-totals: Data Analytics 38/40 · Technical Skills 21/25 · Creativity & Innovation 19/20 · Communication 14/15

Compliance with the General Instructions

Requirement (Gen Instr para)	Status	Remarks
Theme chosen from Para 6	Yes	6.2 Global Conflicts - Impact on Supply Chains
Theme datasets used (Para 5, 8.1)	Yes	ACLED Middle-East aggregates; Sentinel-1 SAR vessel detections
Other open-source datasets used, with citation (Para 5, 7)	Yes	EIA Brent/WTI, Fed H.10 INR/US\$, Energy Institute via OWID, Natural Earth; cited in Annex D
Theme text addressed: AIS dead zones, predict AIS dark spots, identification crisis (Para 6.2)	Yes	Ch 4 (dark zones, prediction), Annex B
Insights and recommendations for India and the Armed Forces (Para 5, 6)	Yes	Ch 7-9; staffed actions
Analysis submitted in PDF (Para 8.2)	Yes	Commander's Edition; full technical report as annex
Screenshots of data analytics software used (Para 8.2)	Partial	Python pipeline and code in Annex C; Power BI screenshots pending
Output files in Power BI (*.pbix) format only (Para 8.2)	NO	Tables, DAX and build guide ready; .pbix to be built
Personal details (Appendix A format) (Para 8.1, 8.2)	NO	Rank, Service No, Unit and Appendix A form pending
Submitted to datathon.ids@gov.in by 30 Sep 26 (Para 8.2)	Pending	-

Actions to close before 30 Sep 26

#	Action	Effect on assessment
1	Build the .pbix in Power BI Desktop from powerbi/ (four pages) and attach it	Mandatory (Para 8.2); +3 to 5.1-5.3
2	Paste two Power BI screenshots into Annex C (Fig C.3, C.4)	Completes Para 8.2 screenshot requirement
3	Fill Rank, Service No, Unit; attach Appendix A details	Mandatory (Para 8.1)
4	Verify the 2026-27 defence budget figure used in the defence-cost comparison	Protects 4.4 (accuracy)
5	Trim the executive summary to five headline numbers	+1 to 7.2

Note: a self-assessment by the author tends to run a few points generous; independent assessors may mark 3-5 points lower.